Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist, *AniPoint Site Research* Vandergrift, PA | May 2026 - Present

- Founded a car wash site selection consultancy (anipointresearch.com) whose feasibility reports (car-count projections, comparable-site benchmarks) supported approval of 2 client sites ($1.5M+ projected yearly revenue).
- Built a revenue driver tree (traffic × capture rate × average ticket) and margin models that turned profit goals into car-count targets.
- Estimated that each new car wash adds $533K a year to its county's car wash revenue, net of cannibalization, using Double ML on a 234-county Census panel (2012–2023) cross-checked with Callaway–Sant'Anna difference-in-differences (within 13%).
- Automated Python pipelines for Monte Carlo risk and low/base/high scenarios (230+ unit tests, data-quality checks).

Senior Data Scientist, *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Developed uplift and segmentation models on millions of HIPAA-protected records, adopted in 90% of campaigns, reducing per-campaign spend by $10K (about 3%) while maintaining baseline enrollment volumes.
- Built an in-house experimentation platform that automated A/B test design and readouts (power analysis, CUPED, significance tests) with Power BI dashboards, saving leadership ~5 hours a week.
- Translated the annual retention survey analysis from R to Python and automated it end to end, cutting turnaround from 2.5 weeks to 1 day.
- Re-architected the churn pipeline on Spark and Databricks (partitioning, caching, removing repeated joins), cutting runtime by 48% and memory by 23%.

Founding Machine Learning Engineer, *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- Replaced a heuristic shipping price rule with a regression model, cutting the median price error vs. actual carrier cost from $30 to $11 (excluding extreme bulk orders) while maintaining an 18% contribution margin.
- Validated the pricing model offline (median absolute error, human-in-the-loop review), then shipped it as a C#/Azure pricing API (10,000+ daily requests).
- Built automated drift monitoring for live inference pipelines and mentored the 16-person engineering team on production ML practices.
- Cut search complaints by 12% and raised search click-through rate from 30% to 35% by tuning Elasticsearch relevance (custom analyzers, field boosting), validated with NDCG@10, human review, and A/B tests.

Data Scientist, *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Raised the offline AUC-ROC of the XGBoost lead-scoring model from 0.62 (prior model) to 0.67 and used SHAP to identify the main drivers (income, age, household composition).
- Confirmed the gain in an A/B test, where model-driven targeting raised gross lead rate (GLR) by 9%.
- Validated GLR out of sample as a 60-day leading indicator of enrollment rate ($R^2 = 0.55$).
- Doubled the precision of high-opportunity zone selection over legacy zoning (15% to 30%, measured against past successful campaigns) using GeoPandas spatial overlays and demographic indexing.
- Led monthly monitoring of up to 10 production models built for clients (accuracy, feature drift, forecast vs. actual) and built a company-wide self-service dashboard on curated Snowflake and Databricks tables.

TECHNICAL PROJECTS

Medicare Claims Intelligence & Fraud Detection medicare-claims-intelligence

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K synthetic CMS claims (DE-SynPUF), with tests that caught 11,253 claims missing a service date.
- Built an out-of-time XGBoost risk model (SHAP, MLflow) whose top-5% list captured 21.0% of next-year high-cost members vs. 14.4% when ranking by prior-year cost (46% more).
- Built ARIMA, ETS, and Prophet claims forecasts that beat a seasonal-naive baseline in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), and validated fraud flags on injected test anomalies (81% precision at review capacity).

EDUCATION

Ph.D. in Mathematics, *University of Pittsburgh* 2017 - 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics, *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics & Planning: KPI Frameworks, KPI / Driver Trees, Standardized Metric Definitions, Leading Indicators, Acquisition & Retention Marketing Analytics, Contribution Margin, Scenario Analysis, Target Setting, Forecasting

Causal Inference & Experimentation: A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling

Modeling & Statistics: Regression, Classification, XGBoost, Time Series Forecasting, Monte Carlo Simulation, SHAP, Drift Monitoring

Data & Tools: SQL (Snowflake, SQL Server), Python (pandas, GeoPandas, statsmodels, scikit-learn, econml, MLflow), R (survey analysis), C#, Databricks, Spark, dbt, Elasticsearch, Azure, Power BI, Excel, Git